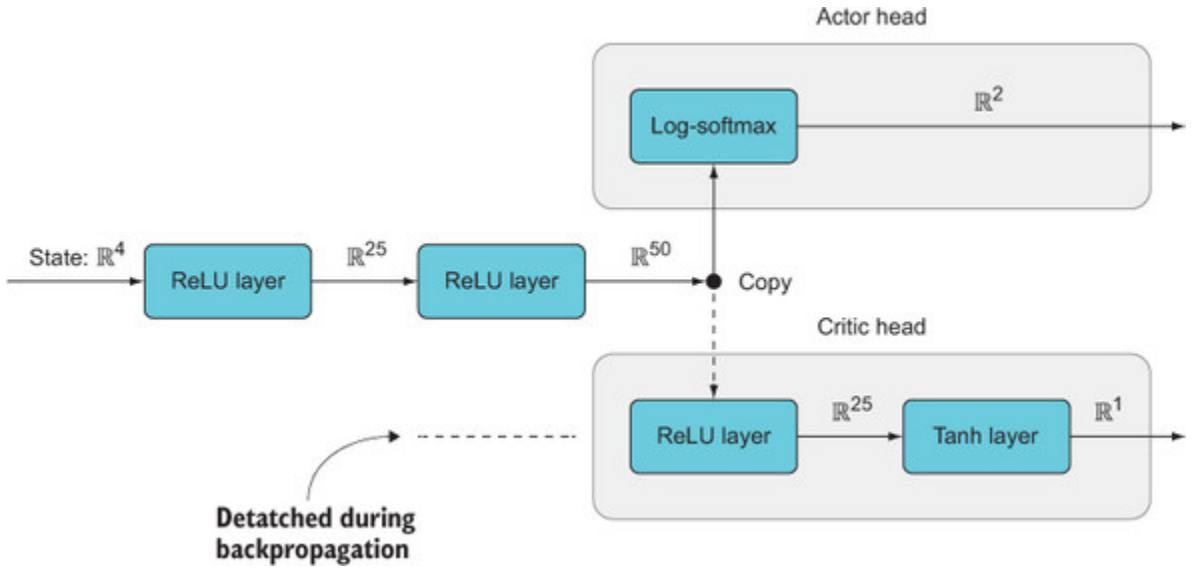
output to the interval (–1,1), which is perfect for CartPole since the rewards are +1 and –1.

Figure 5.11. This is an overview of the architecture for our two-headed actor-critic model. It has two shared linear layers and a branching point where the output of the first two layers is sent to a log-softmax layer of the actor head and also to a ReLU layer of the critic head before finally passing through a tanh layer, which is an activation function that restricts output between –1 and 1. This model returns a 2-tuple of tensors rather than a single tensor. Notice that the critic head is detached (indicated by the dotted line), which means we do not backpropagate from the critic head into the actor head or the beginning of the model. Only the actor backpropagates through the beginning of the model.



In the following listing we develop the code necessary for distributing multiple instances of the actor-critic model across different processes.

Listing 5.5. Distributing the training

```
MasterNode = ActorCritic()
1
MasterNode.share_memory()
2
processes = []
3
params = {
    'epochs':1000,
    'n_workers':7,
}
counter = mp.Value('i',0)
4
for i in range(params['n_workers']):
```